# Ohm’s law

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Connect voltage, current and resistance for an ohmic component.

**SUCCESS CRITERIA - I can...**

1. Identify the voltage across a resistor.
2. Use V=IR with appropriate units.
3. Predict changes while stating what stays fixed.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

Move from r to 3 r away from a point charge. Compare field magnitude and state the source model assumption.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** V = IR; I = V/R.
Ohmic resistor under fixed physical conditions. Resistance can change with temperature.

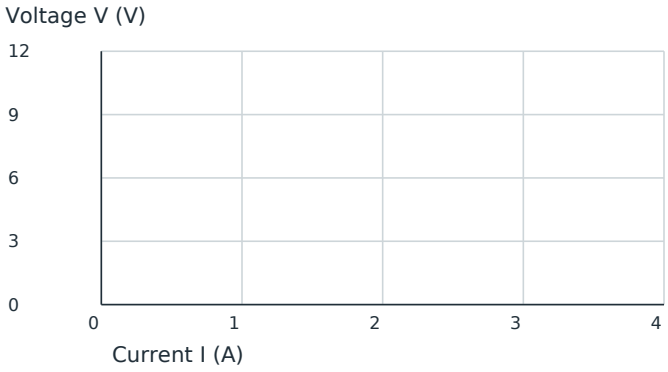
**Worked reasoning:** Across a 4 ohm resistor, a 12 V supply gives I=V/R=3 A. Doubling resistance at the same voltage halves current.

**Your turn:** Draw a one-resistor circuit, label supply voltage and current, and mark both resistor terminals.

**Marzano check - circle:** ① ② ③

## 03 Represent and interpret

Sketch a graph of V against I for a fixed ohmic resistor. What does its slope represent?




**Marzano check - circle:** ① ② ③

## 04 Solve and justify

A resistor has 12 V across it and carries .40 A. Find R. Find current at 9 V if R stays constant.

**Marzano check - circle:** ① ② ③

### 05 Analyze the reasoning

A student doubles resistance while keeping voltage fixed and predicts twice the current. Correct the prediction.

**Marzano check - circle:** 1 2 3

### 06 Choose a method

An EM wave in vacuum has frequency $6\times10^{9}$ Hz. Use $c=3.00\times10^{8}$ m/s. Find wavelength and name the spectral region.

**Marzano check - circle:** 1 2 3

### 07 Exit check

Measured values are (I, V)=(.10, 2.0),(.20, 4.0),(.30, 6.0). Find resistance and explain what supports an ohmic model.

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

An ohmic component has 4 V across it at 0.5 A. Find resistance and current at 10 V. Explain why this extrapolation requires the component to remain ohmic.

**Marzano check - circle:** 1 2 3